

Project Interim Report
On
File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming With Java

SESSION: 2025-26



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Abstract

- The School Administration Management System is a Java-based project developed using Object-Oriented Programming (OOP) concepts such as inheritance and polymorphism. The main purpose of this project is to manage different roles in a school environment, such as teachers and students, in a structured and efficient way.
- In real-world educational institutions, managing data manually can be time-consuming and error-prone. Therefore, this system is designed to simplify the process by using programming concepts to represent real-life entities. It improves efficiency and reduces complexity by organizing information logically.
- The system uses a base class called person, which contains common attributes like name and age. This class is extended into subclasses teacher and student, each having specific properties and behaviors. Method overriding is used to define role-specific actions, and polymorphism is applied to demonstrate dynamic method execution at runtime.

Introduction

Same information about your project but in detail you can also add some of the features of your project. The school administration management system is a Java-oriented programming concepts such as inheritance and polymorphism. It is designed to manage teacher and student records in a simple and efficient way.

This system helps to reduce manual work and provides a structured way to handle information. It is divided into teacher and student classes to demonstrate real-world implementation.

Objectives

- Manage and store teacher and student records efficiently
- Display and track role-based information dynamically
- Reduce manual work and improve accuracy
- Apply Object-Oriented Programming (OOP) concepts such as inheritance and polymorphism
- Improve code reusability and maintainability

Scope:

This project is suitable for small school system to manage teacher and student records in a simple and efficient way. It helps in understanding the practical use of object-oriented programming concepts such as inheritance and Polymorphism.

The system can be further extended by adding features like database integration, graphical user interface (GUI), and additional roles such as admin. It can also be improved to handle larger data and provide more advanced functionalities.

System Requirements:

Hardware: Computer or laptop with minimum 4GB RAM, keyboard, and monitor

Software: Java Development Kit (JDK), any IDE such as VS code or Eclipse, Windows Operating system

Methodology:

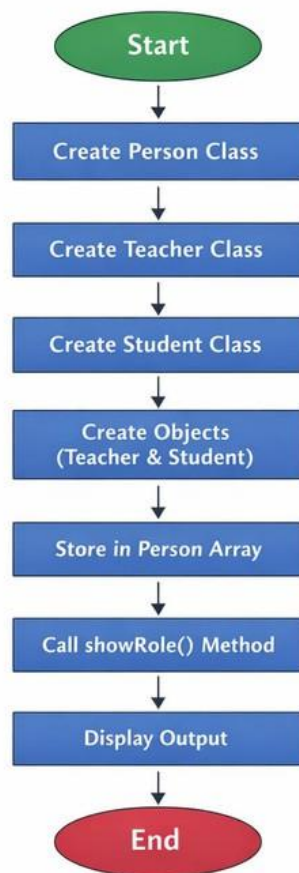
The project works step-by-step to manage teacher and student roles using Object-Oriented Programming Concepts.

1. First, a base class person is created with common attributes like name and age.
2. Then, two subclasses Teacher and student are created by extending the person class.
3. Each subclass has its own properties and overrides the method to define specific behavior.
4. Objects of Teachers are stored in a person array to demonstrate polymorphism.
5. A loop is used to call the method, and the correct output is displayed based on the object type.

Flowchart:

Includes flowchart (Start, Menu, Choice, Operations), and class Diagram representing system structure.
Sample class diagram -

School Administration Management System Flowchart



Person Class

- . Variables: Name, age
- . Methods: ShowRole()

Teacher Class

- Variables: subject, salary
- .Methods: ShowRole()

Student Class

- . Variables: rollNumber, course
- . Methods: showRole()

Implementation:

The project is implemented using three main classes:

Person Class: Stores common details such as name and age. It acts as a base class and includes a method `showRole()` to display a general message.

Teacher Class: Extends the person class and includes additional attributes like subject and salary. It overrides the `showRole()` method to display teacher-related information.

Student Class: Extends the Person class and includes attributes like roll number and course. It overrides the `showRole()` method to display student-related information.

The system uses an array of person objects to store Teacher and Student objects. It includes:

.ShowRole() - Displays role-based information

.Loop traversl() - calls methods dynamically for each object

.The program demonstrates runtime polymorphism by calling the same method for different objects and displaying different outputs

.Sample input / Output

Output:

1. I am a Teacher. I teach students.
2. I am a student. I study subjects.

Future Enhancements

- . Add database for storing records
- . Use GUI for better user interface
- . Add more roles like admin
- . Improve system for handling large data

Gantt Chart:

Task	Duration
Planning (Understanding OOP Concepts)	1 Day
Design Classes (Person, Teacher, Student)	1 Day
Coding (Inheritance & Polymorphism Implementation)	2 Day
Testing (Checking Output & Logic)	1 Day
Documentation (Report Preparation)	1 Day

1. planning & Structure

The project is divided into different tasks such as understanding OOP concepts, designing classes, coding, testing, and documentation. This helps in organizing the developments process in a proper sequence

2. Time Management

The Gantt chart shows the time required for each task. It helps in completing the project within the given time and avoids delays.

Conclusion:

The School Administration Management System Successfully demonstrates the use of object-Oriented Programming concepts such as inheritance and polymorphism. The project helps in understanding how different classes like person, Teacher, and student can be used to represent real-world entities. It shows how method overriding and runtime polymorphism work in a simple Java program. Overall, the project provides a clear idea of code reusability, flexibility, and real-world application of OOP concepts.

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